



ECEN 314 - Signals and Systems

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Module 6 - Impulse Responses

Learning Objectives



- Given a linear, time-invariant system, be able to determine the impulse response of the system.
- Once its impulse response is found, be able to find the output of an LTI system to any input.

Impulse Responses



Previously we found that the output, $y(t)$, of an LTI system due to an input, $x(t)$ is given by a convolution

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(v)h(t-v)dv = \int_{-\infty}^{\infty} h(v)x(t-v)dv.$$

Fundamental in this relationship is the impulse response of the system, $h(t)$, which is the output that will appear when the input is a unit delta function, $\delta(t)$.

But how does one find the impulse response of an LTI system? We'll start with a few examples and then try to generalize as much as possible.

Example 6.1: $y(t) = x(t - t_o) \rightarrow$ System delays the input by t_o .

In this case if the input is $x(t) = \delta(t)$, then the output is $y(t) = h(t) = \delta(t - t_o)$.

Example 6.2: $y(t) = \int_{-\infty}^t x(v)dv \rightarrow$ Output is the integral of the input.

In this case if the input is $x(t) = \delta(t)$, then the output is

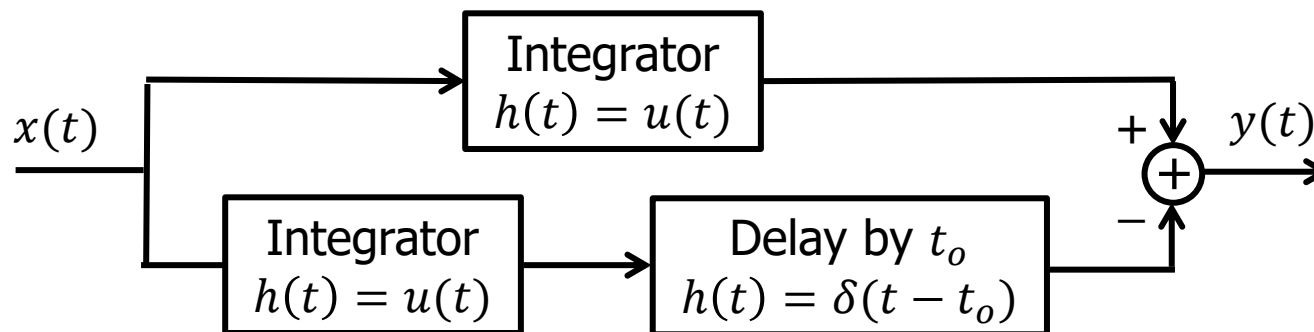
$$y(t) = \int_{-\infty}^t \delta(v)dv = \begin{cases} 1, & t > 0, \\ 0, & t < 0. \end{cases}$$
$$h(t) = u(t).$$

Impulse Responses



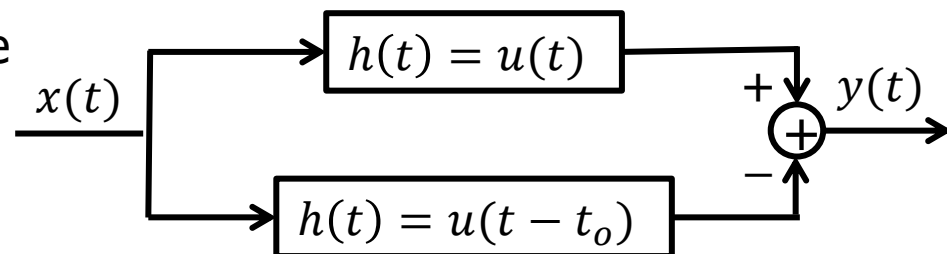
Example 6.3: $y(t) = \int_{t-t_o}^t x(v)dv \rightarrow$ Output is the integral of the input over a window of duration t_o .

Note that in this case: $y(t) = \int_{-\infty}^t x(v)dv - \int_{-\infty}^{t-t_o} x(v)dv$.



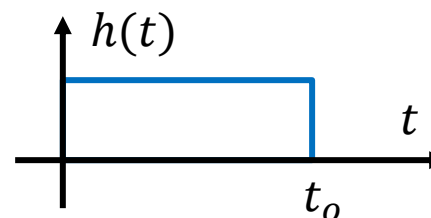
The two systems in series can be replaced by a single equivalent system with impulse response

$$h(t) = u(t) * \delta(t - t_o) = u(t - t_o).$$



Then the two parallel systems can be replaced by a single equivalent system with an impulse response

$$h(t) = u(t) - u(t - t_o) = \text{rect}\left(\frac{t-t_o/2}{t_o}\right).$$



Impulse Responses of a General Class of LTI Systems



It is fairly straightforward to find the impulse response of any LTI described by a differential equation of any order

$$b_n \frac{d^n y}{dt^n} + \cdots + b_2 \frac{d^2 y}{dt^2} + b_1 \frac{dy}{dt} + b_0 y(t) = a_m \frac{d^m x}{dt^m} + \cdots + a_2 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x(t).$$

Taking LTs of both sides and ignoring ICs produces

$$\left(\sum_{k=0}^n b_k s^k \right) Y(s) = \left(\sum_{k=0}^m a_k s^k \right) X(s) \rightarrow Y(s) = \left(\frac{\sum_{k=0}^m a_k s^k}{\sum_{k=0}^n b_k s^k} \right) X(s).$$

Define the polynomials, $b(s) = \sum_{k=0}^n b_k s^k$ and $a(s) = \sum_{k=0}^m a_k s^k$. Then

$$Y(s) = \frac{a(s)}{b(s)} X(s).$$

If the input is an impulse, $x(t) = \delta(t) \rightarrow X(s) = 1$, then the output is the impulse response

$$H(s) = \frac{a(s)}{b(s)} \rightarrow h(t) = \mathcal{L}^{-1} \left\{ \frac{a(s)}{b(s)} \right\}.$$

Using partial fractions and Tables of L.T.'s the impulse response can be found for any system described by a differential equation.

Impulse Responses of LTI Systems



Example 6.4: Find the impulse response of a 2nd order system described by the ODE

$$y''(t) + 3y'(t) + 2y(t) = x(t).$$

Answer: $h(t) = (e^{-t} - e^{-2t})u(t)$

Example 6.5: Find the impulse response of a 2nd order system describe by the ODE

$$y''(t) + 2y'(t) + 2y(t) = x(t).$$

Answer: $h(t) = e^{-t}\sin(t)u(t)$

Finding Outputs of LTI systems



Recall that the input and output of any LTI system are related through a convolution operation involving the impulse response of the system

$$y(t) = \int_{-\infty}^{\infty} x(v)h(t-v)dv.$$

Suppose we take L.T.'s of both sides of this equation

$$\begin{aligned} Y(s) &= \int_0^{\infty} y(t)e^{-st}dt = \int_0^{\infty} \left[\int_{-\infty}^{\infty} x(v)h(t-v)dv \right] e^{-st}dt \\ &= \int_{-\infty}^{\infty} x(v) \left[\int_0^{\infty} h(t-v)e^{-st}dt \right] dv \end{aligned}$$

Using the shifting property of L.T.'s, the inner integral becomes $e^{-sv}H(s)$ so that

$$Y(s) = H(s) \int_{-\infty}^{\infty} x(v)e^{-sv}dv = H(s)X(s)$$

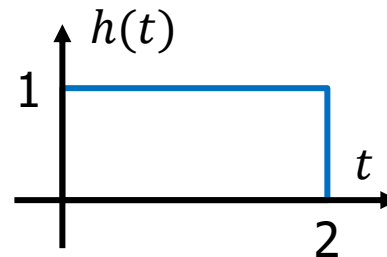
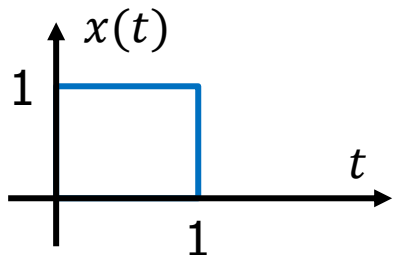
(assuming $x(t) = 0$ for $t < 0$). This suggests an alternative method to compute convolutions using Laplace Transforms:

- ✓ Step 1: Take L.T.'s of $x(t), h(t) \rightarrow X(s), H(s)$
- ✓ Step 2: Multiply to get L.T. of output, $Y(s) = X(s)H(s)$
- ✓ Step 3: Take inverse L.T. of $Y(s)$ to get $y(t)$.

Finding Outputs of LTI systems



Example 6.6: Find $y(t) = x(t) * h(t)$ with $x(t)$ and $h(t)$ as shown below.



Transform Domain Approach:

$$x(t) = u(t) - u(t - 1) \leftrightarrow X(s) = \frac{1}{s} - \frac{1}{s}e^{-s}.$$

$$h(t) = u(t) - u(t - 2) \leftrightarrow H(s) = \frac{1}{s} - \frac{1}{s}e^{-2s}.$$

$$Y(s) = X(s)H(s) = \left(\frac{1}{s} - \frac{1}{s}e^{-s}\right)\left(\frac{1}{s} - \frac{1}{s}e^{-2s}\right) = \frac{1}{s^2}(1 - e^{-s} - e^{-2s} + e^{-3s}).$$

Noting that $\frac{1}{s^2} \leftrightarrow tu(t)$ and using the time shifting property of L.T.'s ($F(s)e^{-st_0} \leftrightarrow f(t - t_0)$) results in

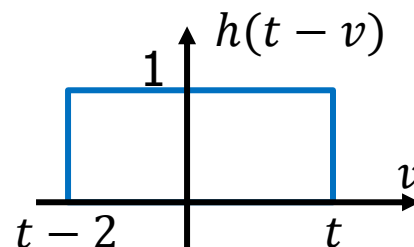
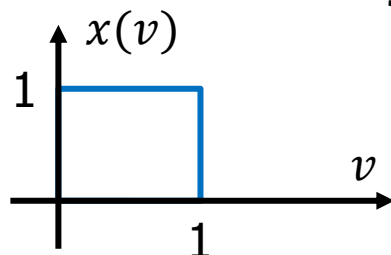
$$y(t) = tu(t) - (t - 1)u(t - 1) - (t - 2)u(t - 2) + (t - 3)u(t - 3).$$

Finding Outputs of LTI systems

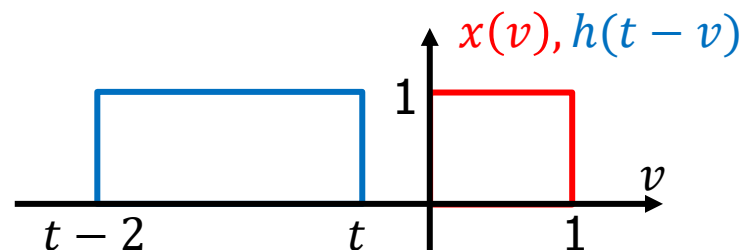


Direct Convolution Approach:

$$y(t) = \int_{-\infty}^{\infty} x(v)h(t-v)dv.$$

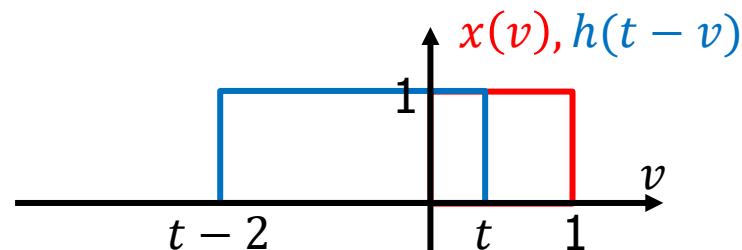


Case 1: $t < 0$, $y(t) = 0$.



Case II: $0 < t < 1$,

$$y(t) = \int_0^t dv = t.$$

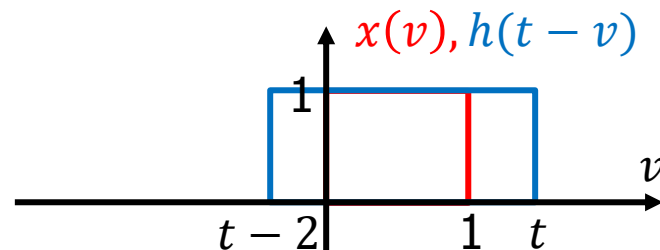


Finding Outputs of LTI systems

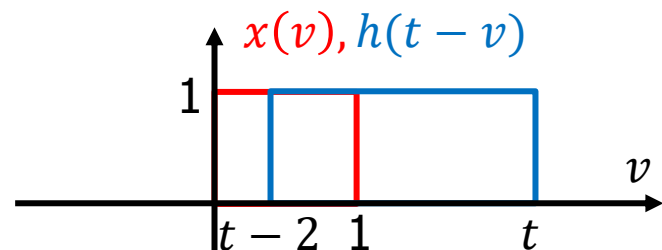


Direct Convolution Approach (continued):

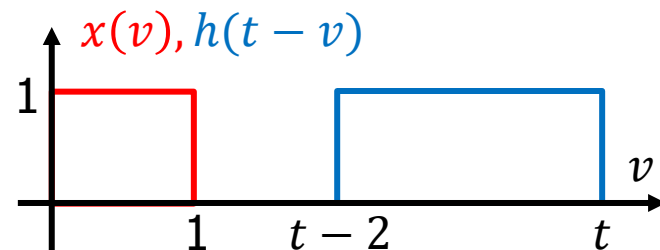
Case III: $1 < t < 2$, $y(t) = \int_0^1 dv = 1$.



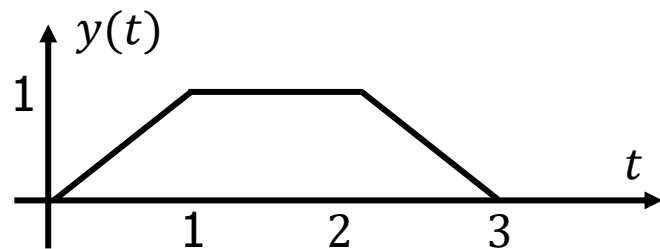
Case IV: $2 < t < 3$,
 $y(t) = \int_{t-2}^1 dv = 3 - t$.



Case V: $t > 3$, $y(t) = 0$.



$$y(t) = \begin{cases} 0, & t < 0, t > 3, \\ t, & 0 < t < 1, \\ 1, & 1 < t < 2, \\ 3 - t, & 2 < t < 3. \end{cases}$$



Finding Outputs of LTI systems



Example 6.7: Find the output of an LTI system described by the differential equation

$$y'(t) + y(t) = x(t)$$

when the input is $x(t) = A\cos(\omega t)u(t)$. What does the output behave like as $t \rightarrow \infty$?

$$\text{Answer: } y(t) = \frac{1}{1+\omega^2} [-e^{-t} + \cos(\omega t) + \omega \sin(\omega t)]u(t)$$

Bonus Material: Impulse Responses of Discrete Time Systems



- One can follow a similar process to find impulse responses of discrete time systems, but a few modifications are needed.
- First off, discrete-time systems are usually described by a difference equation rather than a differential equation
 - ✓ derivatives don't make sense in the context of discrete time.
 - ✓ a difference is the discrete time version of a derivative.

$$\sum_{k=0}^N b_k y[n-k] = \sum_{k=0}^M a_k x[n-k]$$

- Next, rather than using Laplace transforms, we use its discrete-time counterpart known as a z-transform.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Bonus Material: z-transforms of common signals.



Signal, $x[n]$	Transform, $X(z)$	Region of Convergence
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$\alpha^n u[n]$	$\frac{1}{1 - \alpha z^{-1}}$	$ z > \alpha $
$n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$	$ z > \alpha $
$\cos(\omega_o n) u[n]$	$\frac{1 - \cos(\omega_o)z^{-1}}{1 - 2\cos(\omega_o)z^{-1} + z^{-2}}$	$ z > 1$
$\sin(\omega_o n) u[n]$	$\frac{1 - \sin(\omega_o)z^{-1}}{1 - 2\cos(\omega_o)z^{-1} + z^{-2}}$	$ z > 1$
$r^n \cos(\omega_o n) u[n]$	$\frac{1 - r \cos(\omega_o)z^{-1}}{1 - 2r\cos(\omega_o)z^{-1} + r^2 z^{-2}}$	$ z > r$
$r^n \sin(\omega_o n) u[n]$	$\frac{1 - r \sin(\omega_o)z^{-1}}{1 - 2r\cos(\omega_o)z^{-1} + r^2 z^{-2}}$	$ z > r$

Bonus Material: Properties of z-transforms.



Property	Signal, $x[n]$	Transform, $X(z)$
Linearity	$ax[n] + by[n]$	$aX(z) + bY(z)$
Time Shift	$x[n - n_o]$	$z^{-n_o}X(z)$
Time Reversal	$x[-n]$	$X(z^{-1})$
Frequency Shift	$e^{j\omega_o n}x[n]$	$X(ze^{-j\omega_o})$
Frequency Scaling	$\alpha^n x[n]$	$X(z/\alpha)$
Multiplication by n	$nx[n]$	$-z \frac{dX}{dz}$
Cumulative sum	$\sum_{k=-\infty}^n x[k]$	$\frac{X(z)}{1 - z^{-1}}$
Convolution	$x[n] * y[n]$	$X(z)Y(z)$

As with Laplace transforms, we find z-transforms of more complicated signals using transforms of known signals together with various properties.

Bonus Material: Impulse Responses of Discrete Time Systems



- Starting with a generic difference equation

$$\sum_{k=0}^N b_k y[n-k] = \sum_{k=0}^M a_k x[n-k],$$

take z-transforms of both sides of the equation. Using linearity and time-shifting properties we arrive at

$$\sum_{k=0}^N b_k z^{-k} Y(z) = \sum_{k=0}^M a_k z^{-k} X(z).$$

- Dividing both sides by $\sum_{k=0}^N b_k z^{-k}$

$$Y(z) = X(z) \frac{\sum_{k=0}^M a_k z^{-k}}{\sum_{k=0}^N b_k z^{-k}}.$$

- If the input is a delta function, then $X(z) = 1$ and the resulting output (in the z-transform domain) is

$$H(z) = \frac{\sum_{k=0}^M a_k z^{-k}}{\sum_{k=0}^N b_k z^{-k}}.$$

- The impulse response in the time domain could then be found by performing an inverse z-transform (table look-up).

Bonus Material: Impulse Responses of Discrete Time Systems



Example 6.8: Find the impulse response of the discrete-time system described by the difference equation

$$2y[n] + y[n - 1] = x[n]$$

Answer: $h[n] = \frac{1}{2} \cdot \left(-\frac{1}{2}\right)^n u[n]$